

Project Documentation

EduGenie – AI-Powered Learning Assistant

Sample Project Documentation

Project Overview

An AI-powered educational assistant powered by Google Gemini that combines five essential study tools in one platform.

Problem Statement

Challenge	Description
Difficult Concepts	Students need quick, simple explanations for difficult topics.
Self-Testing	Creating practice questions manually takes time.
Long Materials	Summarizing lengthy study material efficiently is difficult.
Unstructured Learning	Students may lack a clear path for learning a topic from scratch.

Technology Stack

Backend: FastAPI (Python) • AI Engine: Google Gemini • Frontend Templates: Jinja2 (HTML) • Server: Uvicorn

Benefits & Impact

- **All-in-One** – Five essential study tools in one assistant.
- **Simple Language** – Responses are tailored for easy, college-level understanding.
- **Time-Saving** – Reduces time spent creating notes, quizzes and study plans.
- **Structured Learning** – Encourages a step-by-step learning approach.
- **Scalable** – Modular backend allows future features to be added.

Conclusion

EduGenie combines Google Gemini with a clean, easy-to-use interface to provide students with a single AI learning companion — from asking questions to building a complete study plan.

Scalability & Future Plan

Scalability

The modular backend (separate Gemini connector, learning-path module and feature routes) allows future features to be added easily.

Future Enhancements

- User accounts & progress tracking
- Quiz scores & learning history
- PDF / notes upload
- Voice-based questions
- Multi-language support